USB-C, ESD

J1 USB-C

VBUS A4 · CC1 A5 · CC2 B5 · D− A7 / B7 · D+ A6 / B6 · SBU1 A8 · SBU2 B8 · SHIELD · GND A1 / S1

VBUS, PWR_FLAG

C1 100n

USB_DM_C, USB_DP_C → U2 USBLC6-2SC6 → USB_DM, USB_DP

R1 5.1k, R2 5.1k

Shield straight to GND

Rd: USB-C sink

3V3 LDO

U3 AP7361C-33FGE

IN, EN, GND, OUT → +3V3

C2 2.2u, C3 10u

277 mA max, 0.62 W at 5.5 V in: EP to GND vias

RP2350A, FLASH, CRYSTAL

100 nF at IOVDD pins 1, 11, 20, 30, 38 (C4–C8), ADC_AVDD 44 and IOVDD 45 (C9, shared as in the vendor layout), USB_OTP_VDD 53 and QSPI_IOVDD 54 (C11). The Decouples field on each names its pin.

Regulator: C14 CIN and C15 COUT (0402 small pads), L1's polarity and copper all follow chips/rp2350.md

C15 COUT at L1; C16 at DVDD pin 23; C17, C18 at DVDD pins 6, 39

L1 3.3u

PWR_FLAG +1V1 → J3 1V1

C15 4.7u, C16 4.7u, C17 100n, C18 100n

C4 100n, C5 100n, C6 100n, C7 100n, C8 100n, C11 100n, C9 100n

R3 33R

C14 4.7u

C13 4.7u

VREG_AVDD, VREG_LX, PWR_FLAG

U1 RP2350A

VREG_AVDD 46 · USB_OTP_VDD 53 · ADC_AVDD 44 · QSPI_IOVDD 54 · IOVDD 1 · VREG_VIN 49 · VREG_LX 48 · VREG_FB 50 · DVDD 6

RUN 26

USB_DM, USB_DP → USB_DM_MCU 51, USB_DP_MCU 52

R4 27R, R5 27R — 27R at the chip

QSPI_SS 60 · QSPI_SCLK 56 · QSPI_SD0 57 · QSPI_SD1 59 · QSPI_SD2 58 · QSPI_SD3 55

U5 W25Q128JVSIQ — CS, CLK, DI/IO0, DO/IO1, WP/IO2, HOLD/RESET/IO3, VCC, GND

C22 100n, C20 15p, C21 15p

Y1 12MHz — XIN 21, XOUT 22, R6 1k

SWCLK 24 · SWDIO 25

GPIO0 2 · GPIO1 3 · GPIO2 4 · GPIO3 5 · GPIO4 7 · GPIO5 8 · GPIO6 9 · GPIO7 10 · GPIO8 12 · GPIO9 13 · GPIO10 14 · GPIO11 15 · GPIO12 16 · GPIO13 17 · GPIO14 18 · GPIO15 19 · GPIO16 27 · GPIO17 28 · GPIO18 29 · GPIO19 31 · GPIO20 32 · GPIO21 33 · GPIO22 34 · GPIO23 35 · GPIO24 36 · GPIO25 37 · GPIO26/ADC0 40 · GPIO27/ADC1 41 · GPIO28/ADC2 42 · GPIO29/ADC3 43

VREG_PGND 47 · GND 61

VREG_PGND joins GND at one point

GPIO HEADERS (TOP VIEW)

USB-C end

J2 — GPIO0 1 … GPIO15 20 — GPIO left

J3 — 20 VBUS … 1 GPIO16 — GPIO right

Pin map: parts/user-io.md. Not Pico pin-compatible: pins 31–39 differ. Headers are not assembled.

SWD, STATUS LED

J4 SWD: SWCLK 1, GND 2, SWDIO 3

R9 1k, D1 RED — GPIO25

RGB LEDS: VLED RAIL, LEVEL SHIFTER, 4 PIXELS

VBUS → D2 SM4007PL → VLED

R11 4.7k, D7 BZT52B5V1, C23 4.7u, C24 100n

PWR_FLAG

C25 100n, C26 100n, C27 100n, C28 100n

U4 SN74LV1T125DBVR — GPIO23 → R12 100k; R10 33R → LED_DIN

D3 WS2812B-2020-V6 — PIX12 → D4 WS2812B-2020-V6 — PIX23 → D5 WS2812B-2020-V6 — PIX34 → D6 WS2812B-2020-V6

VLED 3.5–5.1 V: the rectifier's drop, held up by the 4.7k bleed, keeps the pixels inside 3.3–5.3 V; D7 clamps a hot-plug ring (parts/rgb-leds.md)

BUTTONS

QSPI_SS → R7 1k → SW1 BOOTSEL

RUN → R8 1k → SW2 RESET

Drawn by scripts/schematic.py
Author: Sequoia Alexander
CERN-OHL-P-2.0
RP2350A dev board, 21 x 51 mm
github.com/sequoia-hope/boardroom
Sheet: /
File: easy2350.kicad_sch
Title: easy2350
Size: A3 | Date: 2026-10-23 | **Rev: A**
KiCad E.D.A. 9.0.8 | Id: 1/1